EXAMPLE 2.36

A train was moving at the rate of 54 km h^{-1} when brakes were applied. It came to rest within a distance of 225 m. Calculate the retardation produced in the train.

Solution

The final velocity of the particle $v = 0$

The initial velocity of the particle

$$u = 54 \times \frac{5}{18} m\,s^{-1} = 15\ m\,s^{-1}$$

$$S = 225\ m$$

Retardation is always against the velocity of the particle.

$$v^2 = u^2 - 2aS$$

$$0 = (15)^2 - 2a\,(225)$$

$$450\ a = 225$$

$$a = \frac{225}{450} ms^{-2} = 0.5\, ms^{-2}$$

Hence, retardation = 0.5 $m\ s^{-2}$

2.11 PROJECTILE MOTION

2.11.1 Introduction

When an object is thrown in the air with some initial velocity (NOT just upwards), and then allowed to move under the action of gravity alone, the object is known as a projectile. The path followed by the particle is called its **trajectory.**

Examples of projectile are

1. An object dropped from window of a moving train.
2. A bullet fired from a rifle.
3. A ball thrown in any direction.
4. A javelin or shot put thrown by an athlete.
5. A jet of water issuing from a hole near the bottom of a water tank.

It is found that a projectile moves under the combined effect of two velocities.

i) A uniform velocity in the horizontal direction, which will not change provided there is no air resistance.

ii) **A uniformly changing velocity (i.e., increasing or decreasing) in the vertical direction.**

There are two types of projectile motion:

(i) Projectile given an initial velocity in the horizontal direction (horizontal projection)

(ii) Projectile given an initial velocity at an angle to the horizontal (angular projection)

To study the motion of a projectile, let us assume that,

i) Air resistance is neglected.

ii) The effect due to rotation of Earth and curvature of Earth is negligible.

iii) The acceleration due to gravity is constant in magnitude and direction at all points of the motion of the projectile.

2.11.2 Projectile in horizontal projection

Consider a projectile, say a ball, thrown horizontally with an initial velocity $\vec{u}$ from the top of a tower of height h (Figure 2.39).

As the ball moves, it covers a horizontal distance due to its uniform horizontal

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